

Assuming, for all the above examples,

a= \$s1, b=\$s2, c=\$s3

Type	Structure	Refined Structure
<pre>if(a < b) { c= a+b; } else { c= a-b; } c = a & b;</pre>	So, if (a < b), go to label 1, where label 1 has the code for the case where the condition of if is true. If the condition of if is not true, it should go to else, and the code of else is in label 2, and the code for the rest of the portion below the if else structure remains in the end label. The name of the label can be anything as you wish.	Label 1: c= a+b; Label 2: c= a-b; End: c= a & b; If (a < b) go to label 1 Else if (a > = b) go to label 2

Code	Conversion in MIPS
<pre>if(a < b) { c= a+b; } else { c= a-b; } c = a & b;</pre>	Slt \$t0, \$s1, \$s2 Slt only checks if the first value is greater than the second. Then \$t0=1, else \$t0=0 Beq \$t0, \$zero, label 2 Label 1: add \$s3, \$s1, \$s2 J end Label 2: sub \$s3, \$s1, \$s2 End: and \$s3, \$s1, \$s2
if(a < b)	[Alternative Solution]

<pre> { c= a+b; } else { c= a-b; } c = a & b; </pre>	Slt \$t0, \$s1, \$s2 If you want to jump to a label for \$t0=0, use beq but if you want to jump to a label for \$t1=1 then use bne bne \$t0, \$zero, label 1 Label 2: sub \$s3, \$s1, \$s2 J end Label 1: add \$s3, \$s1, \$s2 End: and \$s3, \$s1, \$s2
<pre> if(a <= b) { c= a+b; } else { c= a-b; } c = a & b; </pre>	Slt \$t0, \$s2, \$s1 If we put \$s2 first, then \$t0=1 if \$s2 < \$s1, (b < a). It doesn't care what is in your condition, it works as a function that only checks whether the first value is smaller than the second. If (b < a) go to label 2 -> \$t0=1 If (b >= a) go to label 1 -> \$t0=0 beq \$t0, \$zero, label 1 Label 2: sub \$s3, \$s1, \$s2 J end Label 1: add \$s3, \$s1, \$s2 End: and \$s3, \$s1, \$s2
<pre> if(a > b) { c= a+b; } else { c= a-b; } c = a & b; </pre>	Slt \$t0, \$s2, \$s1 If (b < a) go to label 1 -> \$t0=1 If (b >= a) go to label 2 -> \$t0=0 bne \$t0, \$zero, label 1 Label 2: sub \$s3, \$s1, \$s2 J end Label 1: add \$s3, \$s1, \$s2 End: and \$s3, \$s1, \$s2

<pre> if(a > b) { c= a+b; } else { c= a-b; } c = a & b; </pre>	[Alternative Solution] <pre> Slt \$t0, \$s2, \$s1 If (b < a) go to label 1 -> \$t0=1 If (b > = a) go to label 2 -> \$t0=0 beq \$t0, \$zero, label 2 Label 1: add \$s3, \$s1, \$s2 J end Label 2: sub \$s3, \$s1, \$s2 End: and \$s3, \$s1, \$s2 </pre>
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[If there are more than one conditionals, eg: if... else if.... Else

The idea is to check whether the condition in if satisfies. If it does, execute block in if code. Else if the condition of if is not true, then go another block of code, that will check whether the condition for else if is true. If condition of else if is true, go to block of code for else if. And if, else if is not true, then go to else block. This process should continue if there are multiple else if as well.

Code	Structure	MIPS Conversion
<pre> if(a > b) { c= a+b; } Else if(a==b) { c= a b; } else { c= a-b; } c = a & b; </pre>	<pre> Label 1: c= a+b; Label 2: c= a b; Label 3: c= a - b; End: c= a & b; If (a > b) go to label 1 Else Check second condition, if a==b, go to label 2 Else, go to label 3. </pre>	<pre> Slt \$t0, \$s2, \$s1 (a first, then b) If (b<a) go to label1, Else check else if condition. Bne \$t0, \$zero, label1 Beq \$s1, \$s2, label2 Label3: sub \$s3, \$s1, \$s2 J end Label 2: or \$s3, \$s1, \$s2 J end Label 1: add \$s3, \$s1, \$s2 End: and \$s3, \$s1, \$s2 </pre>

[if there are more than 1 condition]

Code	MIPS
<pre>if(a< b && b<c) { c= a + b; } else { c= a - b; }</pre>	Check the true or false condition for both cases. Slt \$t0, \$s1, \$s2 Slt \$t1, \$s2, \$s3 and \$t2, \$t0, \$t1 bne \$t2, \$zero, Label 1 Label2: sub \$s3, \$s1, \$s2 J end Label 1: add \$s3, \$s1, \$s2 End: